

Privacy-Sensitive Generative AI Sourcing in Federal Information Systems

Hema Raju Barri¹ and Chandana Charitha Peddinti²

¹ Johns Hopkins University, Baltimore MD, 21218, USA

² University of Maryland, College Park, MD, 20742, USA.

bhemaraju.138@gmail.com, charitha@umd.edu

Abstract

Generative artificial intelligence (GenAI) is becoming part of public-sector information systems, but the sourcing boundary for privacy-sensitive systems remains underexamined. This paper studies whether federal GenAI use cases involving agency-maintained personally identifiable information (PII) are more likely to be purchased from vendors rather than developed internally or through hybrid arrangements. Using the 2025 Federal Agency AI Use Case Inventory, the analysis examines 368 pilot or deployed GenAI use cases across 26 agencies. Vendor-only sourcing appears in 71.3% of PII-involving cases and 38.2% of non-PII cases. In a linear probability model with agency, topic, development-stage, and high-impact controls, PII is associated with a 22.0-percentage-point increase in vendor-only sourcing. The study remains observational and does not infer model openness, hosting, or data-flow location from sourcing mode. The findings contribute use-case-level evidence to AI governance, privacy, public procurement, and digital-government information-systems research.

Keywords: generative AI · information systems · privacy · AI governance · public procurement · digital government

1 Introduction

Governments are moving GenAI from experimentation into operational information systems. These systems do not arrive as isolated models; they are embedded in data pipelines, workflow tools, identity controls, authorization processes, vendor relationships, and accountability structures. That makes GenAI adoption a digital-government governance problem as much as a technical deployment problem. A central but undermeasured governance choice is sourcing: whether an agency builds a system internally, combines internal and contracted resources, or purchases a vendor solution. Privacy makes that sourcing choice theoretically ambiguous. If a GenAI use case involves agency-maintained PII, traditional safeguarding logic suggests that agencies should avoid vendor-only sourcing. Sensitive

data increase contractual incompleteness, monitoring costs, exposure to secondary use, and the cost of mistakes. From this view, privacy-sensitive AI should move inside the organization, or at least toward hybrid development with greater internal control. Yet the opposite may also be true. Operational GenAI requires secure infrastructure, model access, evaluation, data engineering, authorization, monitoring, and specialized staff. Agencies that lack these complementary capabilities may rely more heavily on vendors when privacy stakes are high, purchasing not only software but also compliance and implementation capacity.

This paper asks: among pilot and deployed federal GenAI use cases, how is agency-maintained PII associated with vendor-only sourcing rather than in-house or hybrid development? The question is important for Information Systems (IS) research because privacy-sensitive AI governance is often discussed as a matter of principles, risk frameworks, or model behavior, while sourcing determines who provides controls, who retains technical knowledge, who responds to failures, and who holds bargaining power over future system changes. The empirical setting is the U.S. Office of Management and Budget's newly standardized 2025 Federal Agency AI Use Case Inventory [14]. The inventory reports use-case classification, development stage, whether a pilot or deployed case involves agency-maintained PII, sourcing mode, vendor fields, custom code, authorization status, and risk-related fields. It therefore allows a direct use-case-level test of whether privacy-sensitive GenAI is associated with market procurement or internal development.

The contribution is threefold. First, the paper reframes privacy-sensitive GenAI sourcing as an IS governance boundary problem in digital government. Privacy is not only a data-protection attribute; it can alter the organizational location of the capabilities used to deliver an intelligent system. Second, the paper provides empirical evidence from 368 federal GenAI use cases rather than from adoption intentions, generic disclosures, or ecosystem activity. Third, the paper disciplines a common inference error: sourcing mode does not reveal model architecture. A purchased system may operate in an authorized agency-controlled environment, while an internally developed system may call an external model service. The paper therefore avoids claims about model openness, hosting, or actual data-flow location. The results show a large positive association between PII and vendor-only sourcing. Vendor-only sourcing is reported for 71.3% of PII-involving cases and 38.2% of non-PII cases. After agency fixed effects and use-case controls, PII is associated with a 22.0-percentage-point higher probability of vendor-only sourcing. The estimate is positive in all leave-one-identifying-agency-out exercises. However, the design is observational and the small-cluster inference is not cleanly binary: a restricted wild-cluster bootstrap excludes zero narrowly, while a conservative CR2/Satterthwaite interval includes zero. The evidence is best read as a stable, substantively large association that challenges a simple privacy-internalization narrative, not as an identified treatment effect.

2 Related Work and Conceptual Framework

2.1 GenAI as Public-Sector Information-System Infrastructure

GenAI can improve productivity and diffuse expertise, but its organizational value depends on the information system around the model rather than on the model alone [4, 6]. In public organizations, implementation also involves procurement rules, public accountability, authorization, privacy compliance, workforce capacity, and risk management. Reviews of AI in public administration emphasize the need for explanatory evidence about adoption and governance rather than only conceptual principles or isolated case narratives [11, 22]. IS research is well suited to this problem because it treats technology adoption as sociotechnical. AI capability depends on data, skills,

processes, governance, and complementary resources [8, 12, 19]. In federal GenAI, those complements may be supplied internally, by a vendor, or jointly. Sourcing therefore becomes a visible governance boundary in the public AI stack.

2.2 Privacy, Safeguarding, and Internalization

Transaction-cost economics predicts that market governance becomes less attractive as exchange hazards increase. Sensitive data can make contractual performance difficult to specify, monitoring difficult, and misuse costly to reverse. Privacy research similarly emphasizes imperfect information about data collection, use, and sharing [1, 3]. If PII intensifies contracting hazards, agencies should preserve more internal control over development. This leads to the safeguarding expectation: H1a: PII involvement is negatively associated with vendor-only GenAI sourcing and positively associated with in-house or hybrid development.

2.3 Compliance-Capability Sourcing

A competing view starts from capability scarcity. Classic IS outsourcing research shows that vendor production-cost advantages can outweigh transaction-cost concerns [2] and make-or-buy decisions often combine transaction-cost, knowledge, and measurement explanations [16]. For GenAI, relevant capabilities include secure data integration, identity and access management, evaluation, monitoring, model updates, and documentation. AI-as-a-service also bundles infrastructure and expertise with the application layer [10].

The federal acquisition context makes this capability view plausible. OMB's AI acquisition guidance directs agencies to address privacy, data-use restrictions, testing, and risk management across the acquisition lifecycle [13]. GAO likewise reports that agencies acquire AI through both products and services and need to collect procurement lessons, including contract terms related to data rights and testing [20]. A separate GAO privacy report identifies gaps in federal AI privacy guidance, including privacy-impact assessments, evaluation, auditing, and technological tools for protecting sensitive information [21]. These policy signals do not identify the empirical mechanism, but they show why privacy-sensitive AI sourcing is an urgent digital-government issue.

Under the capability view, privacy may increase demand for vendors with established authorization processes, secure platforms, and implementation expertise. Vendor purchasing then represents a bundle of technical and compliance capacity, not simply a transfer of data outside the agency. This leads to the competing hypothesis:

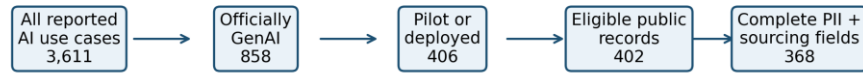
H1b: PII involvement is positively associated with vendor-only GenAI sourcing and negatively associated with in-house or hybrid development. The hypotheses are intentionally competing. The analysis tests which pattern appears in reported federal GenAI use cases, while recognizing that an observational association cannot prove the mechanism.

3 Data and Measures

3.1 Data Source and Sample Construction

The primary source is OMB's 2025 Federal Agency AI Use Case Inventory [14]. Federal agencies are required, subject to specified exceptions, to conduct annual AI inventories and publish machine-readable records. OMB consolidated agency files and standardized categorical responses. The reporting instructions ask agencies to include pre-deployment, pilot, deployed, and recently retired use cases and to report, for pilot and deployed use cases, whether the system involves agency-maintained PII and whether it was purchased from a vendor, developed in-house, or developed with both contracting and internal resources [15].

The public inventory is not the universe of federal AI. The reporting guidance excludes national-security systems, intelligence-community use, Department of War use cases, certain research, and information whose release would conflict with law or policy. Prior work also documents uneven implementation of federal AI-inventory requirements [9]. The estimates therefore apply to publicly reportable, represented, complete 2025 use cases.



Filters: official GenAI classification, pilot/deployed stage, no explicit disclosure-risk/prohibition flag, complete PII and sourcing fields.

Fig. 1. Sample construction from the 2025 Federal Agency AI Use Case Inventory.

The source contains 3,611 individually reported AI use cases, of which 858 are officially classified as GenAI. Restricting to pilot or deployed GenAI leaves 406 cases. Four records explicitly flagged for disclosure risk or prohibition are removed, leaving 402 eligible public cases. Of these, 368 have valid PII and sourcing fields across 26 agencies. The final sample contains 80 PII-involving and 288 non-PII use cases.

3.2 Measures

Table 1 summarizes the core constructs. The main exposure, PII, equals one when the inventory reports that the use case involves PII maintained by the agency. The main outcome, vendor-only sourcing, equals one for "Purchased from a vendor" and zero for in-house or combined contracting and in-house development. A secondary multinomial analysis keeps all three sourcing modes separate. Controls include agency fixed effects, topic-area fixed effects, development stage, and high-impact status. Agency effects absorb stable mission, procurement, capacity, and reporting differences. Topic and stage capture functional and maturity differences. High-impact status is reported separately before and after inclusion because it may be related to both privacy sensitivity and governance choice.

Table 1. Measures used in the INSIGHT submission

Construct	Operational measure	Role in analysis	Interpretation boundary
PII involvement	OMB field indicating agency-maintained PII	Main exposure	Does not capture every sensitive-data type
Vendor-only sourcing	"Purchased from a vendor"	Main outcome	Sourcing, not hosting or model openness
Hybrid sourcing	Contracting plus in-house resources	Secondary outcome	External participation with internal work
In-house development	Agency-developed system	Secondary outcome	Does not prove local deployment
Agency	Reporting agency fixed effects	Main control	Absorbs stable agency-level differences
Topic and stage	Topic area and pilot/deployed status	Main controls	Captures functional and maturity differences
High-impact status	OMB high-impact designation	Full-model control	Potentially downstream of sensitivity
ATO/custom code/vendor	Authorization, code, vendor fields	Mechanism checks	Descriptive; not causal mediation evidence

3.3 Validation

A structured single-reviewer validation audit examined all 80 PII cases and an agency-weighted sample of 80 non-PII cases. Because the audit was conducted by a single reviewer, it should be interpreted as a consistency and error-detection check rather than as a reliability-validated coding exercise. No inter-rater reliability statistics are available. It compared official labels with use-case names, problem statements, outputs, data descriptions, vendor fields, code links, and authorization fields. One record was inconsistent with its official GenAI classification because the description said that the product's AI functions were not used. The main analysis retains OMB's official classification and reports a sensitivity excluding that record. Most importantly, architecture was unobservable in 152 of 160 reviewed cases. The paper therefore does not infer model licensing, hosting mode, local deployment, or external data processing from sourcing fields.

4 Methods

The preferred specification is a linear probability model:

$$\text{VendorOnly}_i = \beta_0 + \beta_{1\text{PII}_i} + \alpha_{a(i)} + \gamma_{t(i)} + \delta\text{Stage}_i + \theta\text{Impact}_i + \varepsilon_i$$

The coefficient beta is the adjusted within-agency difference in vendor-only sourcing probability associated with PII. Although 26 agencies contribute observations, 14 contain both PII classes and directly identify the fixed-effect coefficient; those agencies contain 79 of the 80 PII cases. The estimate is not causal because PII is not randomly assigned and unobserved use-case complexity may influence both privacy status and sourcing. Inference accounts for agency clustering. With only 26 unbalanced clusters and substantial leverage, conventional clustered standard errors can be misleading. The paper reports CR2 bias-reduced standard errors with Satterthwaite degrees of freedom [17] and a restricted wild-cluster bootstrap-t with 9,999 Rademacher replications [5]. Robustness checks include equal-agency weighting, leave-one-agency-out estimates, omission of large agencies one at a time, alternative outcome coding, logistic average marginal effects, a

multinomial model, field-completeness weighting, classification sensitivities, and all-AI interaction checks.

5 Results

5.1 Descriptive Sourcing Patterns

Table 2 shows a pronounced raw sourcing difference. Among non-PII GenAI cases, 26.7% are developed in-house, 35.1% combine internal and contracted resources, and 38.2% are vendor-purchased. Among PII cases, the respective shares are 5.0%, 23.8%, and 71.3%. The raw vendor-only gap is therefore 33.1 percentage points.

Table 2. Sourcing mode by PII involvement

PII involvement	N	In-house	Hybrid	Vendor-only
No	288	77 (26.7%)	101 (35.1%)	110 (38.2%)
Yes	80	4 (5.0%)	19 (23.8%)	57 (71.3%)

The PII group is not otherwise identical. PII cases are more likely to be deployed and more likely to be classified as high impact. These differences motivate the fixed-effect and control specifications.

5.2 Main Estimates

Table 3 reports the PII coefficient as a probability difference. The unadjusted estimate is 0.331. Adding agency fixed effects reduces it to 0.229. Adding stage and topic yields 0.243. The fully adjusted model adds high-impact status and estimates 0.220: conditional on the included factors, a PII-involving GenAI case is associated with a 22.0-percentage-point higher probability of vendor-only sourcing.

Table 3. PII and vendor-only GenAI sourcing

Model	Estimate	CR2 SE	Satt. df	CR2 95% interval	CR2 p-value
Unadjusted	0.331	0.112	5.07	[0.043, 0.618]	0.032
Agency fixed effects	0.229	0.088	6.50	[0.016, 0.441]	0.039
+ Stage and topic	0.243	0.101	6.40	[0.001, 0.486]	0.049
Fully adjusted (+ high impact)	0.220	0.107	6.65	[-0.035, 0.476]	0.080

The fully adjusted restricted wild-cluster-bootstrap interval is [0.007, 0.433], $p=0.044$. The two small-cluster procedures therefore differ narrowly. The appropriate interpretation is a large and stable positive association with meaningful inferential uncertainty, not a binary significance claim.

5.3 Robustness and Sourcing Composition

All 14 leave-one-identifying-agency-out coefficients are positive, ranging from 0.175 to 0.341, with a median near the preferred estimate. Removing DOE, HHS, DOJ, or DHS individually also preserves the positive direction. Equal-agency weighting produces a larger estimate of 0.444, showing that the pooled pattern is not generated solely by agencies with the most observations, although that specification gives small agencies substantial influence. Weighting for observed-field completeness leaves the fully adjusted estimate essentially unchanged.

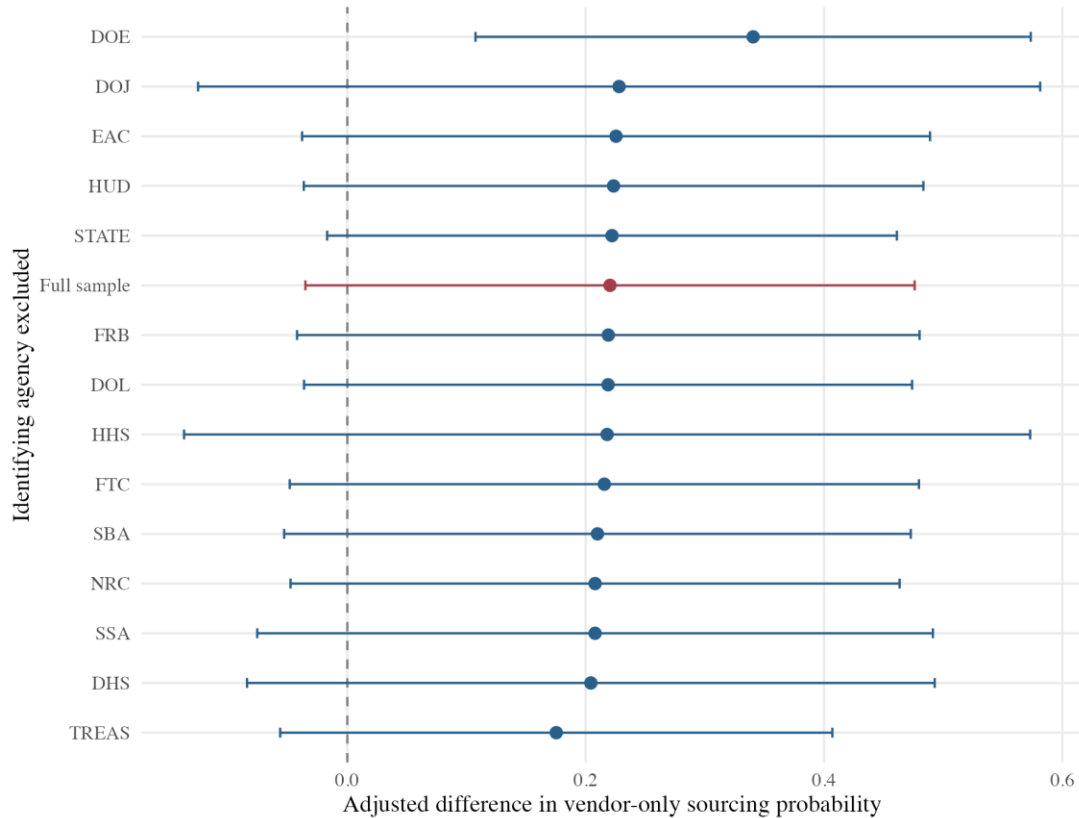


Fig. 2. Fully adjusted estimate and deletions of the 14 agencies that identify the within-agency comparison.

Classification sensitivities do not create the pattern. Removing the single record whose description explicitly denies use of its AI functions yields 0.219. Requiring textual GenAI corroboration retains 210 cases and yields 0.383; that specification is secondary because text availability is selective. A fixed-effect logit model produces an average marginal difference of 0.205, but several cells approach separation. Multinomial predictions associate PII with a 19.7-point increase in vendor-only sourcing, a 10.2-point decrease in hybrid sourcing, and a 9.5-point decrease in in-house development. When hybrid and vendor-only cases are combined into any external participation, the estimate falls to 9.9 points and is imprecise. The main pattern is therefore movement specifically toward vendor purchasing, not merely movement away from purely internal development.

5.4 Mechanism Evidence

The mechanism evidence is intentionally limited. The inventory does not observe internal compliance capacity, procurement routines, mandated enterprise platforms, or system architecture for most cases. Therefore, the analysis cannot determine whether vendor reliance reflects scarce internal capability, procurement convenience, platform standardization, mission complexity, or some combination of these factors. PII is associated with a 10.2-point lower probability of custom code and a 5.8-point higher probability of Authorization to Operate, but both estimates are imprecise. Vendor concentration is also not higher among PII cases: the normalized-vendor Herfindahl index is 0.104 for PII and 0.115 for non-PII cases. Microsoft and Palantir are prominent recognized vendors among PII cases, but vendor fields are incomplete and heterogeneous. These results are consistent with compliance-capability sourcing, but they do not distinguish that explanation from unobserved complexity, platform standardization, agency procurement routines, or mission-specific constraints.

6 Discussion

The results challenge a simple "data cannot leave, therefore build internally" account of privacy-sensitive GenAI governance. In the observed federal GenAI portfolio, PII-involving cases are substantially more likely to be vendor-purchased. This matters for IS research because the boundary around an intelligent system is not only a make-or-buy accounting choice. It shapes data governance, auditability, internal learning, incident response, switching costs, and long-run institutional capacity. Theoretically, the evidence is most consistent with integrated boundary explanations, but the observational design cannot distinguish among the competing mechanisms. Privacy can increase the value of internal control while simultaneously increasing demand for external authorization, security, and maintenance capabilities. Which force dominates likely depends on agency readiness, vendor-market structure, procurement rules, and the maturity of reusable AI platforms.

For digital-government practice, vendor purchasing may accelerate secure deployment, but it can also create dependency. Contracts for privacy-sensitive GenAI should specify data-use restrictions, audit access, subcontractors, deletion, incident reporting, portability, model and system documentation, testing, monitoring, and exit support. Agencies also need sufficient internal expertise to evaluate vendors and govern systems they buy. Public procurement research warns that institutional capability affects due diligence and accountability for AI systems [7], and the NIST AI Risk Management Framework emphasizes risk management across design, deployment, and use [18]. The architecture finding is equally important. The audit found architecture unobservable in 95% of reviewed cases. Vendor-only sourcing should not be treated as evidence of any particular hosting mode, model-access mode, or processing location. Future AI inventories would be more useful if they separated sourcing, hosting environment, model-access mode, data-location rules, and contractual data-use restrictions.

7 Limitations and Future Research

Several limitations are material. First, the design is observational. Fixed effects and controls narrow comparisons but do not create random assignment. Second, reporting is incomplete and excludes important classes of federal AI, including national-security and intelligence use cases. Third, the inventory uses self-reported categories standardized across agencies whose interpretations may differ. Fourth, only 14 agencies directly identify the within-agency fixed-effect coefficient, producing

low effective degrees of freedom despite 26 reported clusters. Fifth, mechanism measures are sparse and do not verify the compliance-capability explanation. Sixth, sourcing does not reveal model openness, local versus hosted deployment, or actual data flows. The next research step is a preregistered out-of-sample replication using the next annual inventory. A stronger design would link use cases to procurement identifiers, USAspending records, contract text, authorization data, and privacy-impact documentation. Interviews with agency AI, privacy, and procurement officials could test whether vendor-only sourcing reflects compliance capability, procurement convenience, system complexity, or mandated platform standardization. Longitudinal data would also distinguish initial adoption from continued operation, retirement, and vendor lock-in. These limitations mean the estimate should be interpreted as a stable association, not as evidence that PII causes agencies to choose vendor-only sourcing.

8 Conclusion

Privacy-sensitive GenAI changes more than risk classification; it can change the organizational boundary through which an intelligent information system is delivered. Across 368 pilot and deployed federal GenAI use cases, PII involvement is associated with a large increase in vendor-only sourcing after agency and use-case controls. The direction is stable across robustness checks, but conservative small-cluster inference and observational selection require caution. The data do not show that vendors improve privacy, that PII leaves the agency, or that purchased systems have any particular model architecture or hosting arrangement. They show something narrower and consequential for Information Systems research: in this federal setting, privacy-sensitive GenAI is reported more often as purchased than internally or jointly developed. Understanding whether that pattern reflects scarce compliance capability, procurement routines, or system complexity is a central question for responsible AI governance in digital government.

Data and code availability. The paper uses public OMB inventory data. Replication scripts, aggregate outputs, validation notes, and checksums are available from the authors on request.

References

1. Acquisti A, Taylor C, Wagman L (2016) The economics of privacy. *Journal of Economic Literature* 54(2):442-492. <https://doi.org/10.1257/jel.54.2.442>
2. Ang S, Straub DW (1998) Production and transaction economies and IS outsourcing: A study of the U.S. banking industry. *MIS Quarterly* 22(4):535-552. <https://doi.org/10.2307/249554>
3. Bélanger F, Crossler RE (2011) Privacy in the digital age: A review of information privacy research in information systems. *MIS Quarterly* 35(4):1017-A36. <https://doi.org/10.2307/41409971>
4. Brynjolfsson E, Li D, Raymond LR (2025) Generative AI at work. *Quarterly Journal of Economics* 140(2):889-942. <https://doi.org/10.1093/qje/qjae044>
5. Cameron AC, Gelbach JB, Miller DL (2008) Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics* 90(3):414-427. <https://doi.org/10.1162/rest.90.3.414>
6. Czarnitzki D, Fernández GP, Rammer C (2023) Artificial intelligence and firm-level productivity. *Journal of Economic Behavior & Organization* 211:188-205. <https://doi.org/10.1016/j.jebo.2023.05.008>
7. Hickok M (2024) Public procurement of artificial intelligence systems: New risks and future proofing. *AI & Society* 39(3):1213-1227. <https://doi.org/10.1007/s00146-022-01572-2>
8. Jöhnk J, Weißert M, Wyrski K (2021) Ready or not, AI comes: An interview study of organizational AI readiness factors. *Business & Information Systems Engineering* 63(1):5-20. <https://doi.org/10.1007/s12599-020-00676-7>
9. Lawrence C, Cui I, Ho D (2023) The bureaucratic challenge to AI governance: An empirical assessment of implementation at U.S. federal agencies. In: *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*, pp 606-652. <https://doi.org/10.1145/3600211.3604701>
10. Lins S, Pandl KD, Teigeler H, Thiebes S, Bayer C, Sunyaev A (2021) Artificial intelligence as a service. *Business & Information Systems Engineering* 63(4):441-456. <https://doi.org/10.1007/s12599-021-00708-w>
11. Madan R, Ashok M (2023) AI adoption and diffusion in public administration: A systematic literature review and future research agenda. *Government Information Quarterly* 40(1):101774. <https://doi.org/10.1016/j.giq.2022.101774>
12. Mikalef P, Gupta M (2021) Artificial intelligence capability: Conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. *Information & Management* 58(3):103434. <https://doi.org/10.1016/j.im.2021.103434>
13. Office of Management and Budget (2025) Driving efficient acquisition of artificial intelligence in government, Memorandum M-25-22. <https://www.whitehouse.gov/wp-content/uploads/2025/02/M-25-22-Driving-Efficient-Acquisition-of-Artificial-Intelligence-in-Government.pdf>
14. Office of Management and Budget (2026a) 2025 Federal Agency AI Use Case Inventory. <https://github.com/ombegov/2025-Federal-Agency-AI-Use-Case-Inventory>
15. Office of Management and Budget (2026b) Guidance on 2025 agency artificial intelligence reporting. [https://github.com/ombegov/2025-Federal-Agency-AI-Use-Case-Inventory/blob/main/Reporting%20Instructions/Guidance%20on%202025%20Agency%20Artificial%20Intelligence%20Reporting%20\(FINAL\).pdf](https://github.com/ombegov/2025-Federal-Agency-AI-Use-Case-Inventory/blob/main/Reporting%20Instructions/Guidance%20on%202025%20Agency%20Artificial%20Intelligence%20Reporting%20(FINAL).pdf)

16. Poppo L, Zenger T (1998) Testing alternative theories of the firm: Transaction cost, knowledge-based, and measurement explanations for make-or-buy decisions in information services. *Strategic Management Journal* 19(9):853-877. [https://doi.org/10.1002/\(SICI\)1097-0266\(199809\)19:9%3C853::AID-SMJ977%3E3.0.CO;2-B](https://doi.org/10.1002/(SICI)1097-0266(199809)19:9%3C853::AID-SMJ977%3E3.0.CO;2-B)
17. Pustejovsky JE, Tipton E (2018) Small-sample methods for cluster-robust variance estimation and hypothesis testing in fixed effects models. *Journal of Business & Economic Statistics* 36(4):672-683. <https://doi.org/10.1080/07350015.2016.1247004>
18. Tabassi E (2023) Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.AI.100-1>
19. Uren V, Edwards JS (2023) Technology readiness and the organizational journey towards AI adoption: An empirical study. *International Journal of Information Management* 68:102588. <https://doi.org/10.1016/j.ijinfomgt.2022.102588>
20. U.S. Government Accountability Office (2026a) Artificial intelligence acquisitions: Agencies should collect and apply lessons learned to improve future procurements, GAO-26-107859. <https://www.gao.gov/products/gao-26-107859>
21. U.S. Government Accountability Office (2026b) Artificial intelligence: OMB action needed to address privacy-related gaps in federal guidance, GAO-26-107681. <https://www.gao.gov/products/gao-26-107681>
22. Zuiderwijk A, Chen YC, Salem F (2021) Implications of the use of artificial intelligence in public governance: A systematic literature review and a research agenda. *Government Information Quarterly* 38(3):101577. <https://doi.org/10.1016/j.giq.2021.101577>